

POWER ELECTRONICS

Switching Power
Supplies

Introduction

- To provide steady, clean voltage
- If load demands more current (discharging filter components) switch to remain *on* longer to compensate
- If input voltage goes up, switch must be turned off sooner each cycle to prevent output voltage from rising

Objectives

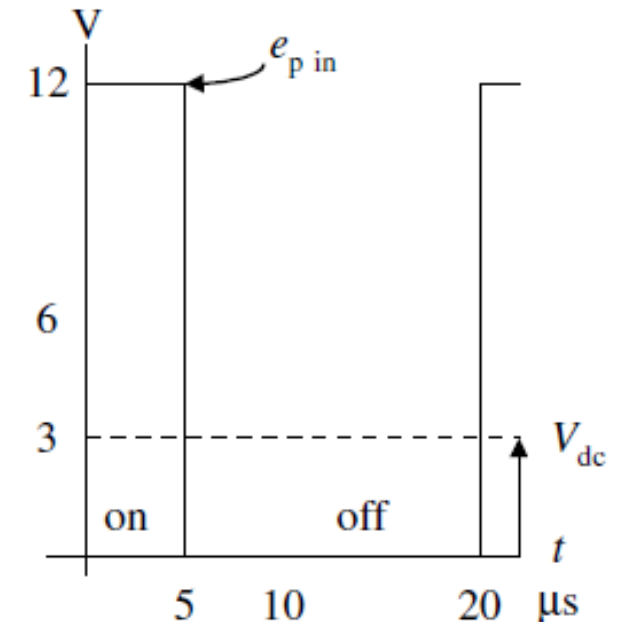
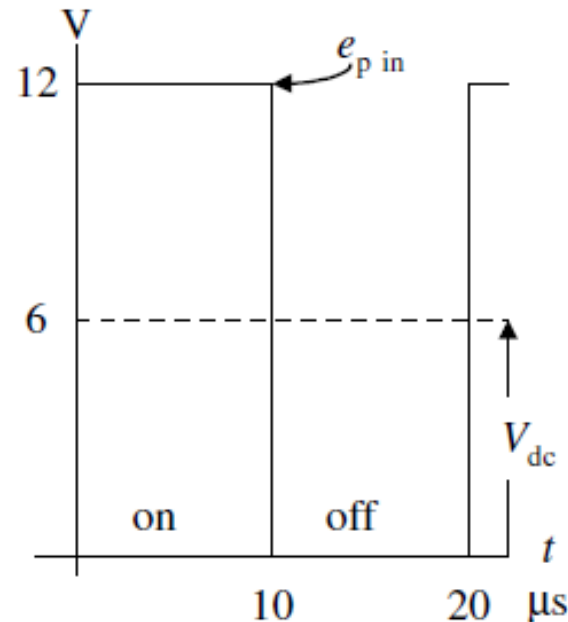
- Analyze a given circuit to determine all waveforms, voltages, currents, and power dissipation for buck, boost and flyback converter
- Design a circuit to meet specifications

Buck Regulators/Converters

- Output is less than the input

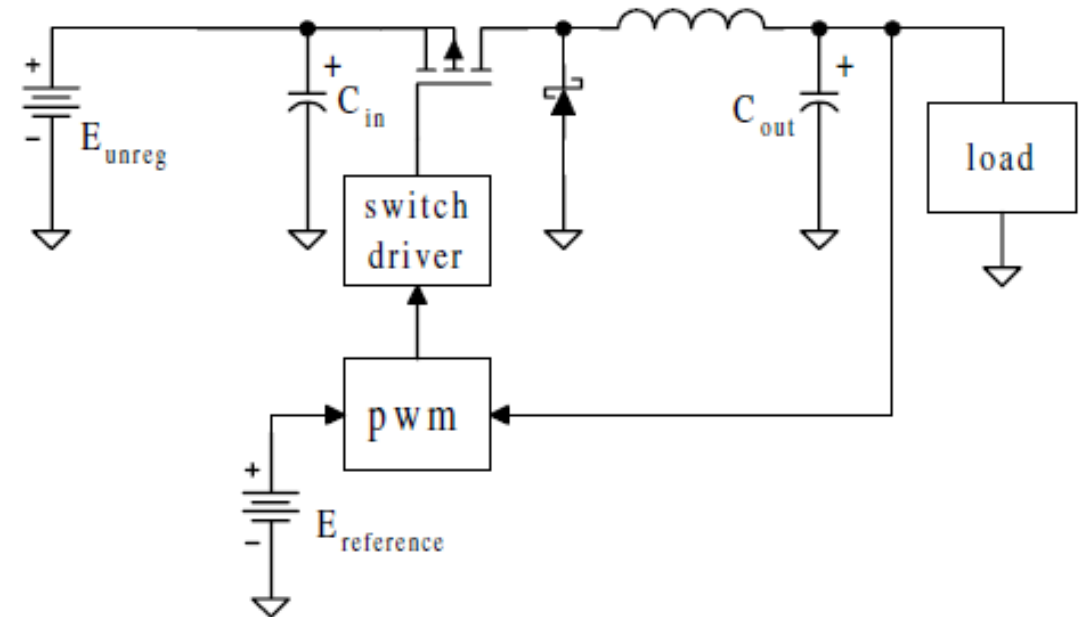
$$V_{DC} = D e_{p\ in}$$

- Calculate the average value for the second signal
- What must be done to keep the output at same value if the input is changed to 7 V?



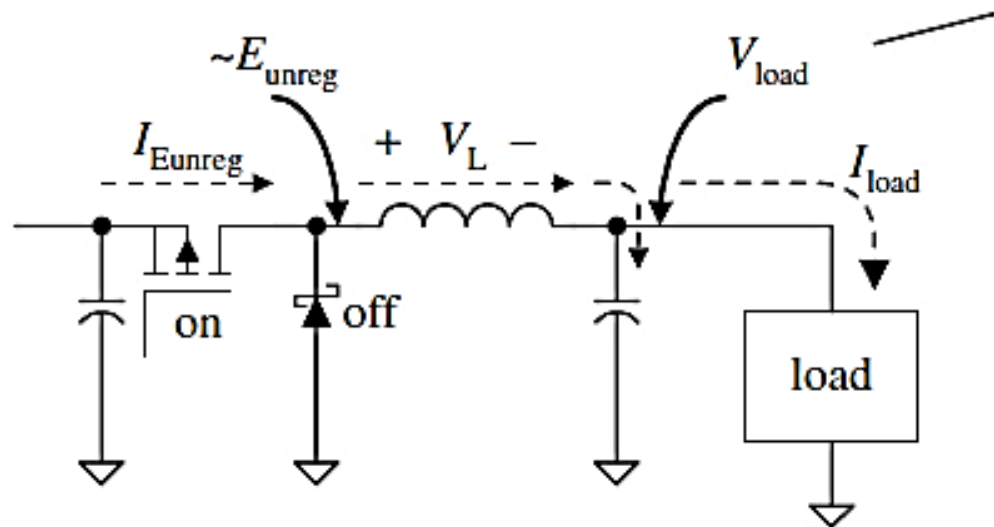
Buck Regulators/Converters

- Alter the load voltage by varying the duty cycle
- Or keep the output voltage constant by correcting the duty cycle in response to variations in input voltage
- Input E_{unreg} may be from raw source, weak grid, a battery, a generator, or a solar array
- $E_{reference}$ provides a stable voltage against which the output is compared
- Comparison is done by PWM
- Output of switch is a rectangle, filter required
- Inductor opposes the change in current
- Capacitor opposes the change in voltage

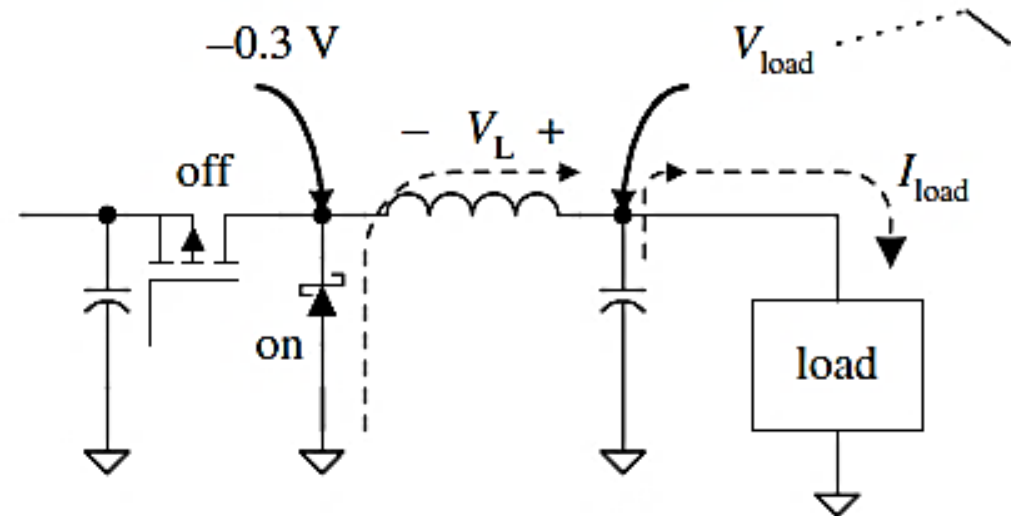


Buck Regulators/Converters

- Charging part of the cycle when the switch is *on*
- Discharging part of the cycle when the switch is *off*



(a) Charge part of the cycle



(b) Discharge part of the cycle

Buck Regulators/Converters

- As diode handles the current of the inductor that is also the load current in discharge cycle so:

$$I_{diode\ continuous} > 1.3 I_{dc\ load\ max}$$

- Diode is turned off in charging cycle, placing E_{unreg} at its cathode. So:

$$PIV > 1.3 E_{unreg\ max}$$

- Inductor voltage equation?
- Inductor voltage?
- Inductor behavior to the changes in current?
- On time?
- Larger the inductor.....??

Buck Regulators/Converters

- A little variation of current actually goes to load
- No need to strictly restrict Δi
- Select $\Delta i \approx 0.25 I_{dc \text{ load}}$ to $0.7 I_{dc \text{ load}}$
- At or below 200 mA dc, allow larger ripple current
- Above 1 A dc, restrict the ripple current to about 25%
- In analysis, ripple current is set by the inductor
- Inductor maximum current $i_{p \text{ inductor}} = I_{dc \text{ load}} + \frac{\Delta i}{2}$

Example

- A buck regulator has an input voltage of 12 V dc, and produces a load voltage of 5 V dc into a $10\ \Omega$ load. Its switch is driven at 100 kHz. The inductor is $220\ \mu\text{H}$. Calculate the following: duty cycle, on time, ripple current, average inductor current, peak inductor current.

Buck Regulators/Converters

- Select input capacitor that has: $I_{rms} C_{in} > 0.5 I_{dc} load$
- For voltage rating: $V_{C_{in}} > 1.5 E_{unreg} max$
- Select capacitor based on Equivalent Series Resistance ESR , current and voltage rating
- Any capacitor above $100 \mu F$ should work

- At output, the capacitor resistance is far smaller than load resistance
- All of the ripple current flows through capacitor
- $\Delta v_{pp} load = \Delta i_{pp} inductor \times ESR_{C_{out}}$

Buck Regulators/Converters

- Output Capacitor should handle the ripple current

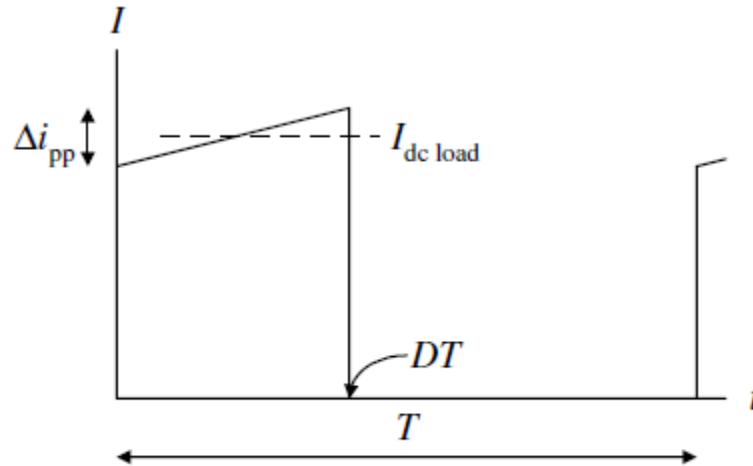
- $i_{rms\ C_{out}} = \frac{\Delta i_{pp\ inductor}}{\sqrt{3}}$

- Select C_{out} as: $C \approx 500 \frac{\mu F}{A} \times I_{dc\ load}$

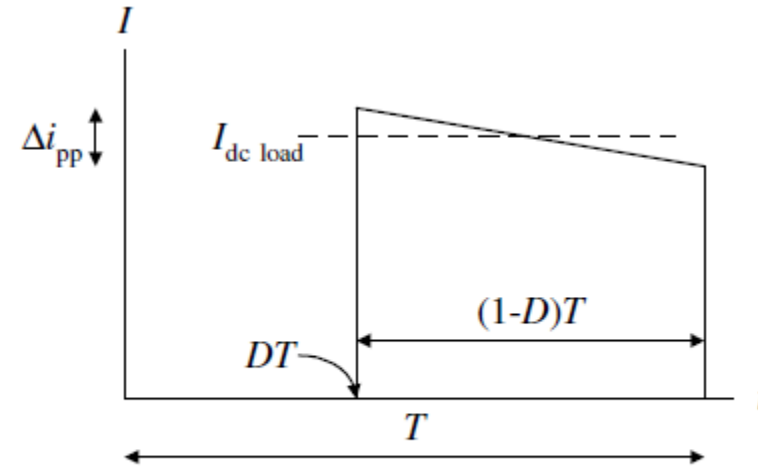
Example

- Calculate the rms value of the current that flows through the capacitor if the peak-to-peak inductor ripple current is 270 mA.

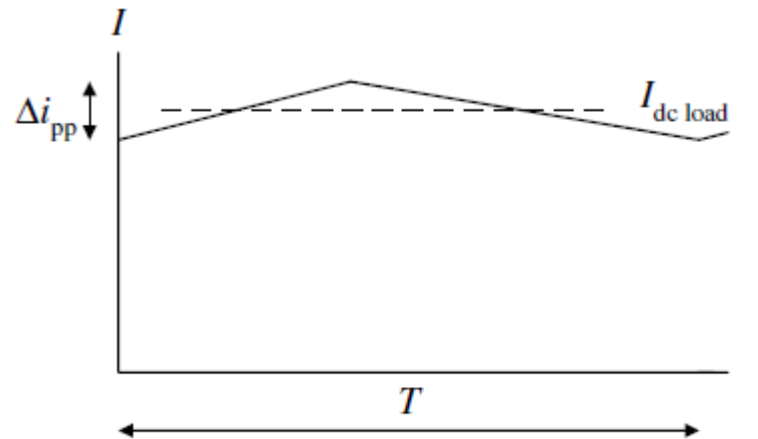
Charging and Discharging Cycle Again



(a) Charge cycle: $i_{Eunreg}, I_Q, i_L, i_{load}$



(b) Discharge cycle: i_L, i_{load}, i_{diode}



(c) Complete cycle: i_L, i_{load}

Example

- Calculate the rms value of the current through the transistor given $I_{dc\ load} = 500\text{ mA}$ and $\Delta i_{pp} = 134\text{ mA}$, $D = 0.42$ complete the calculations twice, first using the trapezoidal equation and secondly assuming that the signal is simple rectangle with $i_p = I_{dc\ load}$. Compare the results.
- Calculate the average value of the same wave using both methods (Homework)

Example

- Calculate the power dissipation of IRF9530 used in buck regulator given $I_{dc\ load} = 500\text{ mA}$ and $\Delta i_{pp} = 134\text{ mA}$, $D = 0.42$ ($R_{on} = 0.6\ \Omega$ at 150°C)
- Also calculate the power dissipation by Schottky diode.
- Calculate the power delivered to the load.
- Calculate the efficiency.

LM2595 Buck Regulator

- Buck regulator IC by National Semiconductor “Simple Switcher” Series.
- $P_Q = DI_{dc\ load}V_{sat\ max}$
- $P_{IC} = I_{quise}E_{unreg}$
- Total Power dissipation $P_{LM2595} = I_{quise}E_{unreg} + DI_{dc\ load}V_{sat\ max}$

Example

- Design a buck converter using a LM2595 to convert an unregulated input voltage of 12 V dc to 5 V dc into 10 Ω . Select all components. Assuming that $T_A = 80^\circ\text{C}$, determine the heat sink needed. Assume the switching frequency is 150 kHz and $V_{ref} = 1.23 \text{ V}$

LM2595 specifications

$$V_{\text{sat max}} = 1 \text{ V}_p$$

$$I_{\text{qules max}} = 10 \text{ mA}_p$$

$$\Theta_{\text{JC}} = 2^\circ\text{C/W}$$

$$\Theta_{\text{JA bare}} = 50^\circ\text{C/W}$$

Reference

Power Electronics, Principles and Applications
J. Michael Jacob

Power Electronics
Ned Mohan